Aarab Ahmad

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Summary

Computer Science undergraduate with strong skills in machine learning, and problem solving. Experienced in building ML-powered web apps, algorithm simulators, and AI-integrated QnA systems. Proficient in Java, Python, React, Flask, and Firebase. Seeking opportunities to apply software engineering and AI expertise to real-world problems.

Education

Lovely Professional University — Jalandhar, Punjab
B. Tech. in Computer Science and Engineering, CGPA: 8.2 — *2023 – Present*

The C.M.S — Basti, U.P
Senior Secondary, Percentage: 90.6 — *2021 – 2023*

Saint. Basil's School — Basti, U.P
High School, Percentage: 91.8 — *2019-2021*

Projects

Charak Samhita QnA | *React, TailWind CSS, Lucide React, FireBase (GitHub)* — Jul 2025 – Present

- Developed an AI-driven QnA platform that delivers cited responses from the Charak Samhita, using React for the frontend and the Google Gemini API for accurate, context-aware answers.
- Designed a responsive, theme-rich UI with Tailwind CSS, incorporating interactive elements such as a feedback modal, "Wisdom of the Day," and a history sidebar, while integrating Firebase for real-time data and secure user sessions.
- Strengthened skills in API consumption, real-time database management, authentication workflows, and building engaging UI/UX experiences with smooth animations and structured user interaction features.

LeakSentry | *Python, Flask, React, HTML5, CSS, SQLite (GitHub)* — Mar 2025 – Present

- Built an end-to-end ML-powered pipeline leakage prediction system designed to process real-time sensor data (pressure, flow, temperature, moisture, acoustic) for accurate leak identification.
- Developed and deployed a full-stack web application featuring interactive sliders, Twilio-based SMS/voice alerts, and a Stripe-enabled subscription workflow.
- Gained hands-on experience in ML model development, full-stack integration, API integration, deployment workflows and building production-ready features such as alerting, payments, and real-time data handling.

Page Replacement Algorithm Simulator | *HTML, CSS, JavScript, Chart.js (GitHub)* — Mar 2025 – Present

- Built an interactive Page Replacement Simulator using PyQt5 and Matplotlib that allows users to input reference strings and frame sizes to evaluate FIFO, LRU, Optimal, Second Chance, and Clock algorithms.
- Added real-time visualizations, including bar-graph comparisons of page faults, step-by-step frame state displays, light/dark themes, input validation, and robust error handling for an intuitive and user-friendly experience.
- Deepened understanding of page replacement strategies, performance analysis, GUI development, and effective visualization techniques by optimizing algorithm comparisons and presenting detailed tabular results.

Achievements

Finalists, LPU's Innotek Annual Innoavtion and Graduating Project Expo

Achieved a contest rating of **1482** in LeetCode Weekly Coding Contests.

Technical Skills

Languages: Java, Python, C/C++, HTML/CSS
Developer Tools: Git, GitHub, Visual Studio Code
Problem Solving: LeetCode, CodeForces, GeeksForGeeks, HackerRank *(Coding Profile)*
Libraries: pandas, NumPy, Matplotlib